

 PT ETANA BIOTECHNOLOGIES INDONESIA	STANDARD OPERATING PROCEDURE	Page : 1 of 6
	TITLE : PERAWATAN TERHADAP MESIN WATER BATH DAIHAN SCIENTIFIC MAINTENANCE OF WATER BATH DAIHAN SCIENTIFIC MACHINE	SOP No. : SOP/EBI/EN-050 Revision : 00 Effective Date : Reviewed Date :
PREPARED BY :	REVIEWED BY :	APPROVED BY :
 WENDI RUKMANSYAH (NON UTILITY ENGINEERING SPV)	 PURNO BUDI KISWANTO (ENGINEERING MANAGER)	 HAPPY MONDA PINTAULI (QA MANAGER)

1. TUJUAN

PURPOSE

Memberikan petunjuk yang jelas mengenai perawatan terhadap Mesin Water Bath Daihan Scientific.

Provide clear instructions on how to maintenance for the Water Bath Daihan Scientific Machine in the production area.

2. RUANG LINGKUP

SCOPE

SOP ini digunakan oleh departement Engineering sebagai pedoman dalam melakukan perawatan Mesin waterbath Daihan Scientific pada area produksi.

This SOP is used by the Engineering department as a guideline for carrying out Machine Waterbath Daihan Scientific maintenance in the production area.

3. TANGGUNG JAWAB

RESPONSIBILITY

3.1. Teknisi Engineering *Preventive Maintenance* bertanggung jawab :

Engineering Technician Preventive Maintenance are responsible :

3.1.1. Melakukan tugas perawatan dengan baik, sesuai dengan SOP ini.

Performing maintenance duties properly, in accordance with this SOP.

3.1.2. Melakukan perawatan sesuai jadwal yang telah ditetapkan.

Perform maintenance according to a predefined schedule.

3.1.3. Melaporkan setiap terjadi kerusakan dan perbaikan yang diperlukan kepada Supervisor Engineering.

Report any damage and repairs that are needed to the Engineering Supervisor.

3.2. Supervisor Engineering bertanggung jawab :

Engineering Supervisor are responsible :

- 3.2.1. Membuat jadwal perawatan secara berkala.
Make schedule of maintenance periodically.
- 3.2.2. Memastikan pelaksanaan perawatan sesuai dengan jadwal.
Ensure implementation of maintenance is according to the schedule.
- 3.2.3. Memastikan semua alat ukur yang terlibat dalam proses perawatan sesuai dengan spesifikasi yang telah ditentukan (dikalibrasi).
Ensure all measuring instruments involved in the maintenance process are in accordance with the specifications specified (calibrated).
- 3.2.4. Mengadakan review bulanan terhadap hasil kerja Teknisi Preventive Maintenance.
Perform a monthly review of the workings of the Preventive Maintenance Technicians.
- 3.2.5. Memberikan pelatihan kepada departemen lain yang terkait.
Provide training to others department which related.
- 3.3. Manajer Engineering bertanggung jawab :
Engineering Manager is responsible :
- 3.3.1. Menentukan *spare parts* pengganti apabila *spare parts* yang sesuai sudah tidak dijual lagi.
Determine replace parts if suitable spare part are no longer sold.
- 3.3.2. Memutuskan dan mengkoordinir tindakan yang diperlukan, apabila perlu dilakukannya penggantian, modifikasi peralatan, perubahan sistem dan menyiapkan prosedur perubahan.
Resolve and coordinate necessary actions, if necessary replacement, equipment modification, system changes and prepared change control.
- 3.4. Operator Mesin Produksi bertanggung Jawab untuk :
The Production Machine Operator is responsible for :
- 3.4.1. Melaporkan jika diareanya ditemukan kelainan-kelainan/kerusakan-kerusakan yang biasanya menghambat dalam pengoperasian mesin tersebut.
Report if the area is detected by defects / abnormalities that normally impede the operation of the machine.
- 3.4.2. Melakukan *daily maintenance* seperti : mengencangkan, klem-klem, ferulle, baut-baut yang kendur, membersihkan *body* mesin / *Main Frame* setiap saat sehingga tidak berkarat dan permukaan mesin tetap baik, membersihkan tiap-tiap mesin yang menjadi tanggung jawabnya, terutama pada bagian dalam dari kerak-kerak yang menempel.
Perform daily maintenance such as: tighten ferulle clamps loose bolts, clean the body of the machine / Main Frame at all times so that it does not rust and the surface of

the machine remains good, cleaned every machine that is the responsibility, especially on the inside of the crust that is attached.

- 3.4.3. Memastikan kondisi alat-alat listrik dan elektronik yang merupakan bagian dari mesin yang ditempatkan pada daerah rawan air, agar selalu terhindar dari percikan air.

Ensure the condition of electrical and electronic equipment that is part of the machine placed in water-prone areas, in order to always avoid splashing water.

- 3.4.4. Melakukan pembuangan air kondensat pada *Air Service Regulator* pada mesin mesin yang dilengkapi dengan sistem pneumatik.

Conduct condensate water discharge at Air Service Regulator on machine equipped with pneumatic system.

- 3.5. Staf Kalibrasi dan Validasi dari Departmen Pemastian Mutu (QA) bertanggung Jawab melakukan kalibrasi terhadap komponen-komponen alat ukur yang digunakan, misalnya Thermometer, Pressure Gauge, Thermo Controller dll, dan melakukan kualifikasi secara rutin.

The Calibration and Validation Staff from Department Quality Assurance is responsible for calibrating the components of the measuring instrument used, eg Thermometer, Pressure Gauge, Thermo Controller etc, and qualify regularly.

4. DEFINISI

DEFINITION

- 4.1. Mesin Water Bath Daihan Scientific adalah alat laboratorium berbentuk wadah yang digunakan untuk memanaskan cairan dengan direndam pada air.

The Daihan Scientific Water Bath Machine is a laboratory tool in the form of a container used to heat liquids by immersing them in water.

- 4.2. *Autonomous maintenance* adalah perawatan mandiri mesin yang dilakukan oleh operator mesin. Bila selama ini operator hanya dilatih untuk mengoperasikan mesin, maka sudah saatnya untuk dilatih lebih lanjut.

Autonomous maintenance is an independent engine maintenance carried out by the machine operator. If all this time the operators are only trained to operate the engine, then it's time to be further trained.

5. PETUNJUK UMUM

GENERAL INSTRUCTION

- 5.1. Sebelum melaksanakan *preventive maintenance*, staf engineering harus berkordinasi dengan operator terkait area tersebut untuk mempersiapkan areanya.
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Before implementing preventive maintenance, engineering staff must coordinate with the operator about the area to prepare the area.

- 5.2. Pada saat pelaksanaan perawatan operator terkait harus mendampingi Staf *Engineering* untuk menghindari kesalahan pengoperasian saat mesin dijalankan atau operator bisa menjalani *autonomous maintenance* pada mesin tersebut.

When carrying out maintenance, the related operator must accompany Engineering to avoid operating errors when the machine is running or the operator can undergo autonomous maintenance on the machine.

- 5.3. Keselamatan kerja harus selalu diperhatikan pada saat *preventive maintenance*, gunakan APD (Alat pelindung diri) yang sesuai peruntukannya pada kondisi saat *maintenance*.

Work safety must be paid attention to at the time of preventive maintenance, use PPE (Personal protective equipment) which according to allotment in condition when maintenance.

- 5.4. Untuk penggantian komponen kritis atau *spare part* yang berdampak langsung dengan produk apabila diganti dengan beda merek dan tipe yang berbeda tetapi satu jenis maka dibuatkan pengendalian perubahan (*change control*).

For the replacement of critical components or spare parts that have a direct impact on the product when replaced with different brands and different types but one type, make change control.

6. ALAT DAN BAHAN

TOOLS AND MATERIALS

6.1. ALAT

TOOLS

1 set tools box

1 set of toolbox

6.2. BAHAN

MATERIALS

6.2.1. Ethanol 20%.

Ethanol 20%.

6.2.2. Kain Wipes..

Wipes Fabric.

7. PROSEDUR

PROCEDURE

Perawatan 3 bulanan (L2) pada Mesin Water Bath Daihan Scientific.

Maintenance 1 month (L2) of Water Bath Daihan Scientific Machine.



Gambar 1. Mesin Water Bath Daihan Scientific

Figure 1. Waterbath Daihan Scientific Machine

- 7.1. Periksa koneksi saklar ON/OFF pada mesin, jika terdapat yg kendur perbaiki atau segera lakukan penggantian.

Check the connection of the ON / OFF switch on the machine, if there is a loose one, fix it or replace it immediately.

- 7.2. Periksa koneksi potensio pada mesin, jika terdapat yg kendur perbaiki atau segera lakukan penggantian,

Check the potential connection on the machine, if something is loose, fix it or replace it immediately.

- 7.3. Periksa kondisi koneksi saklar lainnya, jika terdapat yg kendur perbaiki atau segera lakukan penggantian.

Check the condition of the other switch connections, if something is loose, fix it or replace it immediately.

- 7.4. Periksa koneksi kabel power, kencangkan jika terdapat yg kendur, dan ganti kabel power jika terlihat ada yang rusak.

Check the power cable connections, tighten if any are loose, and replace the power cable if one appears to be damaged.

- 7.5. Bersihkan body mesin jika terdapat yang kotor.

Clean the machine body if there is dirty.

- 7.6. Periksa bagian dalam mesinnya, buka covernya, dan diamati, jika terdapat hal hal yang berpotensi menyebabkan kerusakan segera lakukan perbaikan.

Check the inside of the machine, open the cover, and observe, if there are things that have the potential to cause damage, do repairs immediately.

8. REFERENSI

REFERENCE

Manual book of periodic maintenance Waterbatch Daihan Scientific Machine.

9. LAMPIRAN

ATTACHMENT

Formulir perawatan berkala terhadap Mesin Waterbatch Daihan Scientific List No. SOP/EBI/EN-050-F01A.

Periodically Maintenance form of Machine Waterbatch Daihan Scientific List No. SOP/EBI/EN-050-F01A.

10. RIWAYAT

HISTORY

Tanggal <i>Date</i>	No. Change Control <i>Change Control No.</i>	No. Revisi <i>No. Revision</i>	Alasan Perubahan <i>Reason of Change</i>
	CC-EBI-EN-011-23	00	Dokumen baru <i>New Document</i>

11. DISTRIBUSI

DISTRIBUTION

11.1. Departemen Engineering.

Engineering Department.

11.2. Departemen Pemastian Mutu (QA).

Quality Assurance Department.

11.3. Departemen Produksi.

Production Department.

 PT ETANA BIOTECHNOLOGIES INDONESIA	FORMULIR / FORM PERAWATAN BERKALA PADA MESIN WATER BATH DAIHAN SCIENTIFIC	Document No. : SOP/EBI/EN-050-F01A
	PERIODIC MAINTENANCE OF MACHINE WATER BATH DAIHAN SCIENTIFIC	Effective Date :

Level	:
Month	:
Date	:
Day	:
Hours Meter	:

Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L2											
	Connection of the ON / OFF switch										
	Potential connection										
	Condition of the other switch connections										
	Power cable connections										
	Body machine										
	Check the inside of the machine										

Note :

No	Date	Start	Finish	Duration		Total personnel	Area	Remark
				Hour	Minute			

Remark : C : Cleaning (pembersihan) I : Inspection (pemeriksaan) L : Lubrication (pelumasan) R : Replacement (penggantian) S : Setting (penyetelan) T : Tightened (dikencangkan) W : Welding (pengelasan) G : Good (baik) B : Bad (jelek) * : Critical component	Level : L2 : Every 3 month	Done by : ()	Checked by : ()
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